int

mutex

SharedPtr

recursive_mutex

atomic< StateMachineInternal Action >

map< void *, std::shared _ptr< smacc2::CallbackCounterSemaphore > >

map< std::string, std ::pair< std::function < std::string()>, boost ::any > >

deque< smacc2_msgs ::msg::SmaccTransitionLogEntry >

size_t

SMRunMode

int64_t

SharedPtr

vector< smacc2::ISmaccState * >

shared_ptr< smacc2 ::introspection::SmaccState Info >

SmaccStatus

map< std::string, std ::shared_ptr< smacc2 ::ISmaccOrthogonal > >

shared_ptr< smacc2 ::introspection::SmaccState MachineInfo >

SharedPtr

-reactorWalkDepth_

-transitionLogMutex_

#rth_

-eventQueueMutex_ -m_mutex_

-stateMachineCurrentAction

-stateCallbackConnections

-globalData_

-transitionLogHistory_

-transitionLogHistorySize_

-runMode_

-stateSeqCounter_

#timer_

#currentState_

#currentStateInfo_

#status_msg_

#orthogonals_

#stateMachineInfo_

#eventsLogPub_ #nsmStateMachinePub

#stateMachineStatus Pub_ #transitionHistory Service_ #transitionLogPub_

smacc2::ISmaccStateMachine

+ ISmaccStateMachine()
+ ~ISmaccStateMachine()
+ ISmaccStateMachine()
+ operator=()
+ ISmaccStateMachine()
+ operator=()
+ reset()
+ stop()
+ eStop()
+ joinAsyncClientBehaviorThreads()
+ getOrthogonal()
+ getClientBehavior()
+ getOrthogonals()
+ requiresComponent()
+ requiresComponent()
+ postEvent()
+ postEvent()
+ getGlobalSMData()
+ setGlobalSMData()
+ getStateMachineName()
+ state_machine_visualization()
+ getCurrentStateInfo()
+ publishTransition()
+ onInitialize()
+ onUnhandledStInitialization Failure()
+ getTransitionLogHistory()
+ createSignalConnection()
+ disconnectSmaccSignalObject()
+ disconnectSmaccSignalObject()
+ notifyOnStateEntryStart()
+ notifyOnStateEntryEnd()
+ notifyOnRuntimeConfigured()
+ notifyOnStateExiting()
+ notifyOnStateExited()
+ notifyOnRuntimeConfiguration Finished()
+ getCurrentStateCounter()
+ getCurrentState()
+ getStateMachineInfo()
+ buildStateMachineInfo()
+ getNode()
+ getLogger()
+ getMutex()
# checkStateMachineConsistence()
# initializeROS()
# onInitialized()
# createOrthogonal()
- lockStateMachine()
- unlockStateMachine()
- propagateEventToStateReactors()
- updateStatusMessage()

-mutex_

vector< std::vector < std::shared_ptr< smacc2 ::ISmaccClientBehavior > > >

vector< std::shared _ptr< smacc2::ISmaccClient > >

-clientBehaviors_

#clients_

-stateMachine_

smacc2::ISmaccOrthogonal

+ ISmaccOrthogonal()
+ ~ISmaccOrthogonal()
+ ISmaccOrthogonal()
+ operator=()
+ ISmaccOrthogonal()
+ operator=()
+ setStateMachine()
+ getStateMachine()
+ initState()
+ addClientBehavior()
+ runtimeConfigure()
+ onEntry()
+ onExit()
+ onDispose()
+ getName()
+ requiresComponent()
+ requiresClient()
+ getClients()
+ getClientBehaviors()
+ setGlobalSMData()
+ getGlobalSMData()
+ getClientBehavior()
+ getNode()
+ getLogger()
# onInitialize()
# initializeClients()
# assignClientToOrthogonal()
- currentClientBehaviors()

boost::signals2::signal < Signature, optional _last_value< typename boost ::function_traits< Signature >::result_type >, int, std ::less< int >, function< Signature >, typename extended_signature< function_traits< Signature >::arity, Signature >::function_type, boost:: signals2::mutex >

vector< std::shared _ptr< smacc2::introspection ::ISmaccStateInfo > >

map< smacc2::ComponentKey, std::shared_ptr< smacc2 ::ISmaccComponent > >

-stateMachine_

-pipeFd_

-mutex_

-mutex_

-mutex_

-mutex_

+demangledStateName +fullStateName

+tid_

+staticBehaviorInfo_

+stateReactorsInfo

+eventGeneratorsInfo

+stateMachine_

+parentState_

+transitions_

-children_

-orthogonal_

smacc2::ISmaccClient

+ ISmaccClient()
+ ~ISmaccClient()
+ ISmaccClient()
+ operator=()
+ ISmaccClient()
+ operator=()
+ onInitialize()
+ getName()
+ getComponent()
+ getComponent()
+ getComponent()
+ getType()
+ getStateMachine()
+ connectSignal()
+ requiresClient()
+ getComponents()
+ iterateComponents()
+ postEvent()
+ postEvent()
# onComponentInitialization()
# onStateOrthogonalAllocation()
# createComponent()
# createNamedComponent()
# getNode()
# getLogger()
- initialize()
- setStateMachine()
- setOrthogonal()

#components_

-subscriber_

#stateMachine_

-smaccStateMachine_

-signalDetector_

-subscriber_

-publisher_

-subscriber_

smacc2::introspection ::ISmaccStateInfo

+ SmaccStateInfo()
+ getStateLevel()
+ getStateMachineInfo()
+ getParentState()
+ depth()
+ getAncestors()
+ getFullPath()
+ createChildState()
+ declareTransition()
+ toShortName()
+ getDemangledFullName()
+ getNode()
+ getLogger()

boost::signals2::signal < void(), optional_last _value< typename boost:: function_traits< void()> ::result_type >, int, std ::less< int >, function< void ()>, typename extended_signature < function_traits< void()>::arity, void()>::function_type, boost::signals2::mutex >

smacc2::SmaccSignal < Signature, Combiner, Group, GroupCompare, SlotFunction, ExtendedSlotFunction, Mutex >

vector< std::shared _ptr< smacc2::SmaccEventGenerator > >

shared_ptr< rclcpp ::Logger >

vector< std::shared _ptr< smacc2::StateReactor > >

-stateMachine_

atomic< bool >

bool

uint64_t

vector< smacc2::ISmaccUpdatable * >

vector< std::vector < smacc2::ISmaccUpdatable * > >

SmaccFifoScheduler *

processor_handle

thread

ExecutionModel

double

-mutex_

-command_ -nodeName_

currentOrthogonal

#eventGenerators_

#stateInfo_

#logger_

#stateReactors_

< void()>

#initializationFailed_

#owner_

-initialized_

-refTimestamp_

-end_ -initialized -rosInitialized

-refCosLat_ -refLat_ -refLon_ -refSinLat_

-updatableClients_

-updatableStateElements_

-scheduler_

-processorHandle_

-signalDetectorThread_

-executionModel_

#kDefaultLoopRateHz #kInitializePubRateHz -loop_rate_hz

-publisher_

-publisher_

smacc2::ISmaccState

+ ISmaccState()
+ ~ISmaccState()
+ ISmaccState()
+ operator=()
+ ISmaccState()
+ operator=()
+ getStateMachine()
+ getName()
+ getParentState()
+ getNode()
+ getLogger()
+ initializationFailed()
+ getClassName()
+ configure()
+ requiresComponent()
+ requiresComponent()
+ requiresClient()
+ getGlobalSMData()
+ setGlobalSMData()
+ createStateReactor()
+ createStateReactor()
+ createEventGenerator()
+ postEvent()
+ postEvent()
+ notifyTransition()
+ notifyTransitionFromTransition TypeInfo()
+ getStateReactors()
+ getEventGenerators()
+ getOrthogonal()
+ getClientBehavior()
+ getEventGenerator()
+ getStateReactor()

smacc2::ISmaccComponent

+ ISmaccComponent()
+ ~ISmaccComponent()
+ ISmaccComponent()
+ operator=()
+ ISmaccComponent()
+ operator=()
+ getName()
# onInitialize()
# onComponentInitialization()
# postEvent()
# postEvent()
# onStateOrthogonalAllocation()
# requiresComponent()
# requiresComponent()
# requiresClient()
# createSiblingComponent()
# createSiblingNamedComponent()
# getNode()
# getLogger()
# getStateMachine()
- initialize()
- setStateMachine()

smacc2::SignalDetector

+ SignalDetector()
+ initialize()
+ setProcessorHandle()
+ runThread()
+ join()
+ stop()
+ terminateScheduler()
+ joinAsyncClientBehaviorThreads()
+ pollingLoop()
+ postEvent()
+ queueWorkItem()
+ getNode()
+ getLogger()
+ notifyStateConfigured()
+ notifyStateExited()
+ notifyStateInitialization Failed()
+ notifyRosInitialized()
+ pollOnce()
- findUpdatableClientsAndComponents()
- findUpdatableStateElements()

-refLat_ -refLon_

thread

pid_t

-launched_ -shutdownRequested_

function< void(const std::string &)>

#parentState_

smacc2::SmaccSignal < void()>

-attitudeInvalid_ -hasMessage_ -localAltitudeInvalid

-localPositionInvalid_

uint8_t

float

-firstMessage_ -headingGoodForControl

-valid_ -velocityValid

-xyGlobal_ -zGlobal_

cl_px4_mr::MapProjection

+ MapProjection()
+ MapProjection()
+ initReference()
+ isInitialized()
+ referenceTimestamp()
+ referenceLat()
+ referenceLon()
+ project()
+ reproject()

optional< rclcpp::Time >

optional< rclcpp::Duration >

-refTimestamp_

-readerThread_

-agentPid_

-isShutdownRequested_

-currentState

-postInitializationFailure EventFn_

smacc2::ISmaccClientBehavior

+ ISmaccClientBehavior()
+ ~ISmaccClientBehavior()
+ ISmaccClientBehavior()
+ operator=()
+ ISmaccClientBehavior()
+ operator=()
+ getStateMachine()
+ getName()
+ requiresClient()
+ requiresComponent()
+ requiresComponent()
+ onEntry()
+ onExit()
+ executeOnEntry()
+ executeOnExit()
# runtimeConfigure()
# postEvent()
# postEvent()
# getCurrentState()
# postInitializationFailure Event()
# dispose()
# getNode()
# getLogger()
- onStateOrthogonalAllocation()

+onArmed_ +onDisarmed_ +onLanded_ +onModeChanged_

+onLocalPositionReset

+onPositionReceived_

-armingState_ -navState_ -prevArmingState

-prevNavState_

-goalActive_

-hasPublished_

-xyResetCounter_ -zResetCounter_

-heading_ -refAlt_ -x_ -y_ -z_

projection_

-lastUpdate_

-periodDuration_

cl_px4_mr::CpVehicleLocal Position

+ CpVehicleLocalPosition()
+ ~CpVehicleLocalPosition()
+ onInitialize()
+ getX()
+ getY()
+ getZ()
+ getHeading()
+ isValid()
+ isHeadingGoodForControl()
+ isVelocityValid()
+ globalRefValid()
+ getRefLat()
+ getRefLon()
+ getRefAlt()
+ getRefTimestamp()
+ getXyResetCounter()
+ getZResetCounter()
+ projectToNed()
+ reprojectFromNed()
- onPositionMessage()

-goalX_ -goalY_ -goalZ_ -xyTolerance

-zTolerance_

TrajectorySetpoint

-enabled_

smacc2::ISmaccUpdatable

+ ISmaccUpdatable()
+ ISmaccUpdatable()
+ ~ISmaccUpdatable()
+ ISmaccUpdatable()
+ operator=()
+ ISmaccUpdatable()
+ operator=()
+ executeUpdate()
+ setUpdatePeriod()
# update()

-statusPub_

-timeoutSec_

cl_px4_mr::CpMicroRosAgent

+ CpMicroRosAgent()
+ ~CpMicroRosAgent()
+ launch()
+ shutdown()
+ isLaunched()
+ getPid()
+ getNodeName()
- killProcessesRecursive()

function< void()>

optional< std::future < int > >

-onFailure_ -onFinished_ -onSuccess_

cl_px4_mr::CpVehicleStatus

+ CpVehicleStatus()
+ ~CpVehicleStatus()
+ onInitialize()
+ isArmed()
+ isLanded()
+ getNavState()
+ getArmingState()
- onStatusMessage()

+onGoalReached_

cl_px4_mr::CpFailsafeFlags

+ CpFailsafeFlags()
+ ~CpFailsafeFlags()
+ onInitialize()
+ hasMessage()
+ isAltitudeInvalid()
+ isLocalAltitudeInvalid()
+ isLocalPositionInvalid()
+ isHealthOk()
- onFailsafeMessage()

cl_px4_mr::CpVehicleCommand

+ CpVehicleCommand()
+ ~CpVehicleCommand()
+ onInitialize()
+ sendCommand()
+ arm()
+ forceArm()
+ disarm()
+ setOffboardMode()
+ land()
+ takeoff()

-localPosition_

-localPosition_

-lastSetpoint_

-postFailureEventFn

-postFinishEventFn

-postSuccessEventFn_

-onEntryThread_ -onExitThread_

smacc2::SmaccAsyncClient Behavior

+ onStateOrthogonalAllocation()
+ SmaccAsyncClientBehavior()
+ ~SmaccAsyncClientBehavior()
+ SmaccAsyncClientBehavior()
+ operator=()
+ SmaccAsyncClientBehavior()
+ operator=()
+ onSuccess()
+ onFinished()
+ onFailure()
+ requestForceFinish()
+ executeOnEntry()
+ executeOnExit()
+ waitOnEntryThread()
+ onSuccess()
+ onFinished()
+ onFailure()
# postSuccessEvent()
# postFailureEvent()
# dispose()
# isShutdownRequested()
- waitFutureIfNotFinished()

cl_px4_mr::CpGoalChecker

+ CpGoalChecker()
+ ~CpGoalChecker()
+ onInitialize()
+ update()
+ setGoal()
+ clearGoal()
+ isGoalActive()

cl_px4_mr::CpTrajectorySetpoint

+ CpTrajectorySetpoint()
+ ~CpTrajectorySetpoint()
+ onInitialize()
+ setPositionNED()
+ hold()
+ republishLast()
+ getLastSetpoint()

-completed_

-microRosAgent_

-failsafeFlags_

#vehicleStatus_

#vehicleCommand_

#goalChecker_

#trajectorySetpoint_

#localPosition_

-trajectorySetpoint_

cl_px4_mr::CpOffboardKeep Alive

+ CpOffboardKeepAlive()
+ ~CpOffboardKeepAlive()
+ onInitialize()
+ update()
+ enable()
+ disable()
+ isEnabled()

atomic< int64_t >

optional< std::chrono ::steady_clock::time _point >

-timeoutMs_

-watchdogStart_

-offboardKeepAlive_

cl_px4_mr::CbPx4ClientBehaviorBase

+ onStateOrthogonalAllocation()
+ ~CbPx4ClientBehaviorBase()
+ setTimeout()
+ hasTimeout()
+ update()
# wireCompletionSignals()
# postPx4Success()
# postPx4Failure()

Rate

-rate_

cl_px4_mr::CbConnectMicro RosAgent

+ CbConnectMicroRosAgent()
+ onEntry()
+ onStateOrthogonalAllocation()